

**Executive Master Program:
Financial Engineering**

Master Thesis

High-Frequency Market Making via Reinforcement Learning under Different Volatility Regimes

Student	Onur Emiroglu
Home Address	<i>Fener Mahallesi, 1994. Sk, 07160, Muratpasa</i>
Country	<i>Turkey</i>
Institution	Karlsruhe Institute of Technology
Supervisor	Prof. Dr. Maxim Ulrich
Advisor / Co-supervisor	Ardalan Azarnejad
Matriculation No.	2643340
Date of submission	24.06.2026

Declaration

Student	Onur Emiroglu
Home Address	Fener Mahallesi, 1994. Sk, 07160, Muratpasa
Country	Turkey
Date of birth / location	07.10.1994 / Istanbul
Number of matriculation	2643340

I sincerely affirm to have composed this thesis work autonomously, to have indicated completely and accurately all aids and sources used and to have marked anything taken from other works, with or without changes. Furthermore, I affirm to have observed the constitution of the KIT for the safeguarding of good scientific practice, as amended.

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Confidentiality Statement

No confidentiality statement is required for this thesis.

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Abstract

This thesis studies whether explicit volatility-regime information improves a Proximal Policy Optimization (PPO)-based high-frequency market-making (HFMM) agent in a controlled synthetic limit-order-book (LOB) environment. The simulator uses an arithmetic Brownian-motion mid-price, a sticky Markov volatility-regime process, rolling realized-volatility detection, and Poisson-arrival fills. The PPO agent chooses bid and ask quotes through a discrete half-spread and skew parameterization, and it is compared with a fixed-spread strategy and an Avellaneda-Stoikov analytical baseline.

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once $\sigma_{\hat{}}$ is already observed by the PPO policy. The results are consistent with signal redundancy. The evidence comes from the canonical experiments: out-of-sample evaluation, detector robustness, oracle-label ablations, regime-conditional reward shaping, mild model misspecification, and a signal-informativeness sweep. The strongest interpretation is conditional and synthetic-market bounded: the explicit regime label mostly repackages volatility information already present in the continuous signal. The thesis makes no live-trading deployment claim and does not claim to prove the internal PPO mechanism.

Keywords:

market making; reinforcement learning; PPO; volatility regimes; signal redundancy; limit order book; equivalence testing

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List of Abbreviations / Symbols

Symbol	Meaning
S_t	Mid-price at time step t .
q_t	Inventory after step t .
X_t	Cash account after step t .
W_t	Marked-to-market wealth or equity, $W_t = X_t + q_t S_t$.
$\sigma_{\hat{t}}$	Rolling realized-volatility estimate observed by the PPO policy.
R_t	Reward at time t .
h	Quoted half-spread in ticks.
m	Quote skew in ticks.
A, k	Poisson fill-intensity scale and decay parameters.
η	Inventory penalty coefficient.
γ	Avellaneda-Stoikov inventory-risk aversion parameter.
τ	Remaining horizon.
$\lambda(\delta)$	Poisson fill intensity at quote distance δ .
$\delta, \delta_{bid}, \delta_{ask}$	Quote distance variables.
$\rho_t(\theta)$	PPO probability ratio.
$A_{\hat{t}t}$	Advantage estimate.
ϵ_t	Standard-normal innovation in the mid-price process.
z_t	Latent volatility regime.
P_{ij}	Markov transition probability.

Abbreviation	Meaning
ABM	Arithmetic Brownian motion.
AS	Avellaneda-Stoikov baseline.
HFMM	High-frequency market making.
OOS	Out-of-sample evaluation.
PPO	Proximal Policy Optimization.
RV	Realized volatility.
TOST	Two One-Sided Tests equivalence procedure.
WP	Work package.

Term	Definition
regime-aware PPO	PPO policy variant that receives a categorical regime one-hot channel.
regime-blind PPO	PPO policy variant that omits the regime one-hot but still observes $\sigma_{\hat{t}}$.
σ_{only}	Ablation variant using the continuous volatility signal without categorical regime labels.
combined	Ablation variant using $\sigma_{\hat{t}}$ and estimated categorical regime labels.
oracle_full	Ablation variant using $\sigma_{\hat{t}}$ and the true regime label.
regime_only	Ablation variant using estimated categorical regime labels without $\sigma_{\hat{t}}$.
oracle_pure	Ablation variant using the true regime label without $\sigma_{\hat{t}}$.

1 Introduction

1.1 Problem Statement and Motivation

Market makers continuously choose bid and ask quotes while balancing spread capture against inventory risk. When volatility changes, the same quote width can become either too passive or too exposed. A natural response is to provide a learning agent with regime labels such as low, medium, and high volatility. The central question is whether that categorical label adds decision-relevant information when the policy already observes a continuous volatility estimate.

1.2 Research Objective

The objective is to test the incremental value of explicit categorical volatility-regime labels in PPO market making under a controlled synthetic environment. The comparison is intentionally information-design focused: the regime label is evaluated against a continuous $\sigma_{\hat{}}$ signal, not against an observation space with no volatility information.

1.3 Contributions

1. A controlled HFMM simulator with Poisson-arrival fills, fees, latency, and Markov volatility regimes.
2. A Gymnasium environment in which PPO controls quote half-spread and skew.
3. An OOS comparison of fixed-spread, Avellaneda-Stoikov, regime-aware PPO, and regime-blind PPO strategies.
4. A five-variant ablation separating continuous volatility information from estimated and oracle regime labels.
5. Detector, reward-shaping, misspecification, and signal-informativeness checks that bound the interpretation.

1.4 Scope and Limitations

The thesis is a controlled synthetic-market study. It does not claim production deployment viability, real-order-book external validity, or a proven PPO representation mechanism. The result is conditional on the simulator, signal design, PPO hyperparameters, and degradation calibration tested here.

1.5 Thesis Structure

Section 2 reviews the state of the art. Section 3 presents the methodology, including the mathematical model, use-case scenario, system architecture, algorithmic protocol, and experimental setup. Section 4 reports the results. Section 5 evaluates and discusses the evidence. Section 6 concludes with the thesis outlook.

2 State of the Art

2.1 Background and Related Work

The literature relevant to this thesis falls into four connected strands: classical inventory-aware market making, reinforcement-learning market making, volatility regimes and non-stationarity, and statistical interpretation of small or practically negligible differences. The purpose of the review is not to claim a new analytical quoting rule or a new reinforcement-learning algorithm. It is to locate a narrower information-design question inside an established market-making literature.

Classical market-making models provide the inventory-risk and quote-placement foundation. A market maker earns spread income by posting bid and ask quotes, but stochastic execution creates inventory exposure whose value changes with the mid-price. Avellaneda and Stoikov (2008) formalize this tradeoff through a reservation price and spread term that depend on inventory, risk aversion, volatility, remaining horizon, and order-arrival intensity. Guéant, Lehalle and Fernandez-Tapia (2013) extend this inventory-control view and provide tractable approximations for optimal quotes under inventory constraints.

This analytical literature motivates the thesis design in two direct ways: it explains why inventory, volatility, quote distance, and fill intensity are central state variables, and it supplies the Avellaneda-Stoikov baseline used as a non-learning comparator. Related preprints by Fodra and Labadie (2012, 2013) are useful additional context for inventory-constrained and state-dependent market-making models, but they are treated cautiously as preprints. The thesis does not attempt to derive a new closed-form quoting rule; it uses the analytical tradition to frame the simulation and benchmark design.

A second strand replaces closed-form rules with learned policies. Spooner et al. (2018) demonstrate that reinforcement learning can learn market-making behavior in a simulated limit-order-book setting and show why reward design and inventory control matter. Spooner and Savani (2020) extend the discussion toward robustness under adversarially varied market conditions. These studies support the use of simulator-based evaluation for learned quoting policies, while also showing that market-making performance can be sensitive to the assumptions used to generate training and evaluation episodes.

Deep reinforcement-learning studies with market signals are closest to the present information-design question. Gašperov and Kostanjčar (2021) study market making with signals through deep reinforcement learning; Gašperov et al. (2021) review reinforcement-learning approaches to optimal market making; and Gašperov and

Kostanjčar (2022) train agents in a richer Hawkes-process limit-order-book simulator. This literature motivates learned quoting, state representation, reward shaping, robustness checks, and simulator-based evidence. The present thesis is narrower: PPO is used as a controlled testbed for whether one additional volatility-regime channel has incremental value, not as a claim of algorithmic optimality.

Volatility regimes and non-stationarity are economically meaningful. A high-volatility state can increase inventory risk, change fill-risk tradeoffs, and make a quote width that was appropriate in a calm period too aggressive in a turbulent period. The thesis therefore models a latent three-state Markov volatility process and a causal rolling realized-volatility detector. This design should not be read as a claim that volatility regimes are irrelevant. The tested question is more specific: whether categorical regime labels add incremental value once the policy already observes a continuous realized-volatility estimate.

The key state-representation issue is redundancy. The continuous signal $\sigma_{\hat{}}$ gives the PPO policy a graded estimate of recent realized volatility. A low/medium/high label derived from related volatility information may help if it summarizes latent structure that $\sigma_{\hat{}}$ does not expose, but it may add little if the continuous signal already contains the useful variation for quote placement. This gives the exact gap addressed by the thesis: When a PPO market-making policy already observes a continuous realized-volatility estimate, does adding an explicit categorical volatility-regime label improve out-of-sample performance in a controlled synthetic HFMM environment?

Finally, the empirical interpretation requires care because the thesis evaluates an incremental signal, not the existence of volatility effects in general. Lakens (2017) and Lakens, Scheel and Isager (2018) motivate equivalence testing as a way to distinguish a merely non-significant difference from evidence that a difference is small relative to a pre-specified practical bound. The TOST framing is therefore used as background for the later statistics section; it does not by itself prove that labels contain no information.

3 Approach of Thesis / Methodology

3.1 Mathematical Formulation

Synthetic Mid-Price Process

$$S_{t+1} = S_t + \sigma_t \sqrt{dt} \varepsilon_t \quad (1)$$

Here S_t is the mid-price, σ_t is the step volatility in ticks or price units according to the simulator configuration, dt is the time step, and ε_t is an independent standard-normal shock.

Markov Volatility-Regime Process

$$P(z_{t+1} = j \mid z_t = i) = P_{ij} \quad (2)$$

Here z_t is the latent volatility regime at step t , L/M/H denote low, medium, and high volatility, and P_{ij} is the sticky transition probability from regime i to regime j .

$$P = \begin{bmatrix} 0.9967 & 0.0023 & 0.0010 \\ 0.0042 & 0.9917 & 0.0041 \\ 0.0010 & 0.0030 & 0.9960 \end{bmatrix} \quad (2a)$$

Rows correspond to the current regime L/M/H, and columns correspond to the next regime L/M/H. The large diagonal probabilities make the regimes persistent, or sticky, so volatility states tend to last for many steps rather than switching erratically.

This persistence matters because rolling realized-volatility detection is only meaningful when regime-dependent volatility differences last long enough to be observed through a finite window.

The regime controls the volatility multiplier applied to the base sigma parameter. The canonical full experiments use sigma multipliers [0.6, 1.0, 1.8] for L, M, and H.

Realized Volatility Signal and Regime Detection

$$\hat{\sigma}_t = \sqrt{[(1/w) \sum_{i=t-w+1}^t (\Delta S_i)^2]} \quad (3)$$

Here $\hat{\sigma}_t$ is the rolling realized-volatility signal, w is the rolling window length, ΔS_i is the mid-price increment over step i , and the summation runs over the most recent w increments.

Estimated regime labels are assigned by thresholding $\hat{\sigma}_t$ after a warmup period. The main reported training and evaluation pipelines use the causal `rv_baseline`

detector. The `rv_dwell` detector is retained only as an auxiliary/offline robustness comparison, while the HMM detector is an additional robustness variant.

Illustrative Synthetic Environment Path

As shown in Figure 3.1, the synthetic mid-price path, the policy's rolling `sigma_hat` signal, and the latent low-, medium-, and high-volatility regime sequence evolve together around the dashed chronological train/test split. The figure matters because it distinguishes the continuous signal observed by the policy from the latent regime used for controlled evaluation, while keeping the evidence bounded to the synthetic HFMM environment.

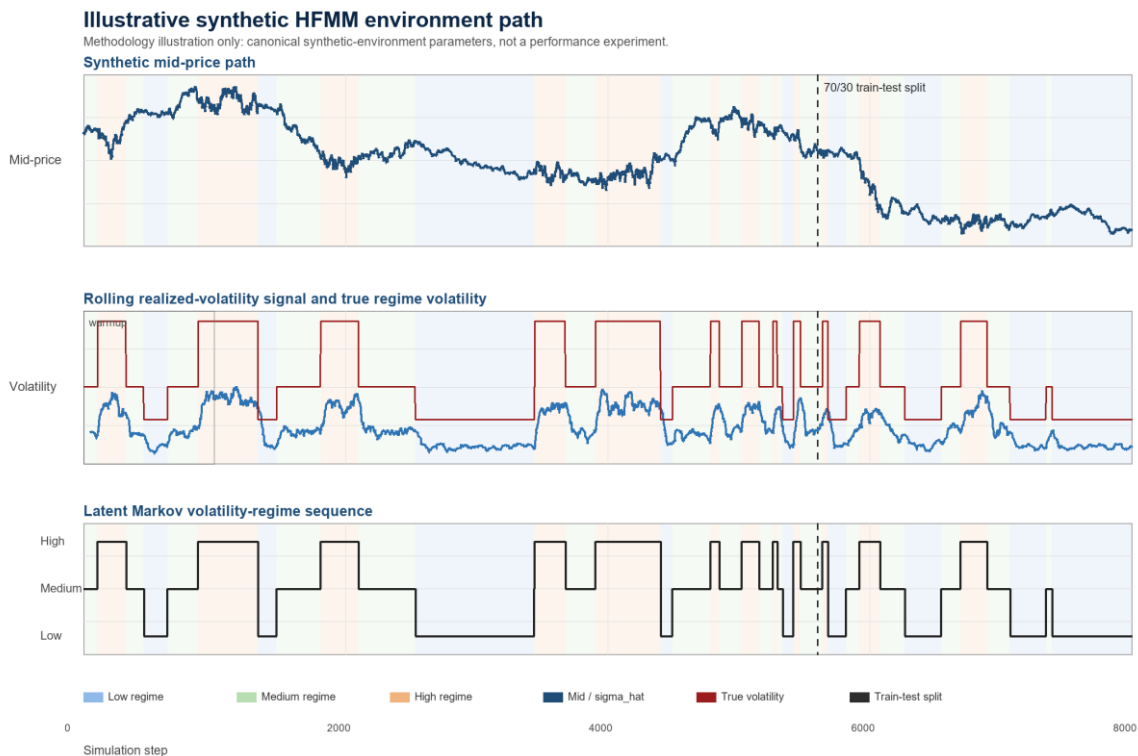


Figure 3.1. Illustrative synthetic HFMM environment path. The upper panel shows the generated mid-price path. The middle panel compares the rolling realized-volatility signal observed by the policy with the true regime-dependent volatility level. The lower panel shows the latent Markov volatility-regime sequence. The dashed vertical line marks the chronological 70/30 train-test split used for out-of-sample evaluation. This figure is a methodology illustration, not a regenerated performance artifact.

Limit-Order Fill Model

$$\lambda(\delta) = A \exp(-k\delta) \quad (4)$$

Here $\lambda(\delta)$ is the fill intensity at quote distance δ , A is the baseline arrival scale, and k controls the exponential decay as quotes move farther from the mid-price.

Intuitively, quotes closer to the mid-price fill more often because they are more attractive to incoming market orders. Quotes farther away earn more spread if they fill, but the exponential intensity makes those fills less likely.

$$P(\text{fill} \mid \delta) = 1 - \exp[-\lambda(\delta) dt] \quad (5)$$

Here $P(\text{fill} \mid \delta)$ is the per-step fill probability and dt is the simulator step length.

Quote Parameterization

$$h = h_{idx} + 1, \quad m = m_{idx} - 2 \quad (6)$$

Here h_{idx} and m_{idx} are the two discrete PPO action components, h is the half-spread in ticks, and m is the quote skew in ticks.

The half-spread h controls quote width and aggressiveness: small h is more aggressive and large h is more passive. The skew m shifts the bid and ask distances asymmetrically, allowing the policy to lean against inventory or other state pressure while still quoting both sides.

$$\delta_{bid} = \max(1, h + m), \quad \delta_{ask} = \max(1, h - m) \quad (7)$$

Here δ_{bid} and δ_{ask} are the bid and ask quote distances in ticks. The max operator enforces a minimum quote distance of one tick.

Inventory, Cash, Equity, and Reward

$$q_{t+1} = q_t + F_{bid_t} - F_{ask_t} \quad (8)$$

Here q_t is inventory, F_t^{bid} is the bid-side fill indicator or count, and F_t^{ask} is the ask-side fill indicator or count.

$$X_{t+1} = X_t - F_{bid_t} P_{bid_t} + F_{ask_t} P_{ask_t} - \text{fees}_t \quad (9)$$

Here X_t is cash, P_t^{bid} and P_t^{ask} are the executed bid and ask prices, and fees_t denotes transaction costs.

$$W_t = X_t + q_t S_t \quad (10)$$

Here W_t is mark-to-market wealth or equity, X_t is cash, q_t is inventory, and S_t is the mid-price.

The equity measure is marked to the mid-price for consistency across simulated strategies. This is an evaluation convention inside the controlled simulator and should not be interpreted as immediately liquidatable wealth in a real limit-order book. In practice, liquidating inventory would generally require crossing the spread, facing queue priority and available depth, and potentially incurring additional adverse-selection or market-impact costs. These effects are part of the real-market boundary discussed later, not part of the current controlled experiment.

$$R_t = W_{t+1} - W_t - \eta q_{t+1}^2 \quad (11)$$

Here R_t is the reward and η is the inventory-penalty coefficient. Fees are included in the cash update and are not counted a second time in the reward.

The ηq^2 term discourages the agent from earning apparent PnL by accumulating large directional inventory. It connects the learning objective to market-making risk control rather than pure speculation on the mid-price path.

Avellaneda-Stoikov Baseline

$$r_t = S_t - q_t \gamma \sigma^2 \tau \quad (12)$$

Here r_t is the reservation price, γ is inventory-risk aversion, σ is volatility, and τ is remaining horizon.

$$\delta_{AS} = 0.5 \gamma \sigma^2 \tau + (1/\gamma) \ln(1 + \gamma/k) \quad (13)$$

Here δ_{AS} is the Avellaneda-Stoikov half-spread and k is the same fill-intensity decay parameter used in the execution model. The implemented deltas are clipped to configured minimum and maximum bounds.

PPO Objective

The clipped PPO objective follows Schulman et al. (2017); it is included here as the training-objective reference for the PPO implementation, not as a new algorithmic contribution.

$$\rho_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta,old}(a_t | s_t) \quad (14)$$

Here $\rho_t(\theta)$ is the PPO probability ratio, π_θ is the current policy, $\pi_{\theta,old}$ is the behavior policy used to collect the rollout, a_t is the action, and s_t is the state.

$$L^{CLIP}(\theta) = E_t[\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t)] \quad (15)$$

Here θ denotes policy parameters, $\rho_t(\theta)$ is the PPO probability ratio, $\hat{A}_t$ is the advantage estimate, and ϵ is the clipping parameter. The thesis uses this only as a concise training-objective reference.

PPO is used because the thesis needs a stable, widely used policy-gradient algorithm for stochastic sequential control, not because it claims PPO is the best possible market-making RL algorithm. The action is discrete but two-dimensional MultiDiscrete: the agent chooses half-spread h and skew m , which together form a 25-action quote grid. DQN would require value-based handling of this grid and is less natural for the policy-gradient control framing used here; SAC is more naturally aligned with continuous actions and would expand the scope; and A2C is simpler but often less

stable or sample-efficient than PPO. PPO therefore provides a reliable baseline algorithm for controlled ablation of the observation channels.

3.2 Synthetic HFMM Use-Case Scenario

Synthetic HFMM Scenario

The use case is a synthetic market maker posting one bid and one ask quote at each step. The environment supplies mid-price dynamics, volatility estimates, and regime labels; the agent receives stochastic fills and is evaluated on wealth, risk-adjusted performance, inventory tails, and fill behavior.

Trading Agents and Strategy Variants

Table 3.1 Strategy variants

Strategy or variant	Observed information	Purpose
Naive	No learned state dependence	Fixed-spread baseline.
Avellaneda-Stoikov	Inventory, volatility, horizon	Analytical inventory-risk baseline.
ppo_aware	$\sigma_{\hat{}}$ and estimated regime one-hot	Original regime-aware PPO.
ppo_blind	$\sigma_{\hat{}}$ without regime one-hot	Original regime-blind PPO.
sigma_only	Continuous $\sigma_{\hat{}}$ only	Tests whether $\sigma_{\hat{}}$ alone carries the signal.
combined	$\sigma_{\hat{}}$ and estimated regime one-hot	Tests estimated categorical incremental value.
oracle_full	$\sigma_{\hat{}}$ and true regime one-hot	Tests perfect-label incremental value.
regime_only oracle_pure	/ Categorical labels without $\sigma_{\hat{}}$	Anchor label-only variants.

As seen in Table 3.1, the design separates the naive and Avellaneda-Stoikov baselines from the original PPO-aware/blind comparison and from the sigma_only, combined, and oracle-label ablations. This separation makes each comparison answer a distinct information-design question rather than treating all strategies as interchangeable benchmarks.

Information-Design Question

The information-design question is: when a PPO market-making policy already observes a continuous realized-volatility estimate, does adding an explicit categorical volatility-regime label improve out-of-sample performance in a controlled synthetic HFMM environment?

The canonical evidence answers this question negatively in the tested setting. Explicit categorical volatility-regime labels do not provide robust incremental value once

$\sigma_{\hat{}}$ is already observed by the PPO policy. The results are consistent with signal redundancy.

3.3 System Architecture

Run Lifecycle and Reproducibility Layer

`run.py` dispatches jobs from JSON configuration files, creates a timestamped run directory, snapshots the config, records git metadata, writes logs and metrics, and finalizes run status. Long WP6 jobs support resume validation against the saved config snapshot.

Synthetic Market and Detector Layer

`src/wp1/sim.py` implements the fill and inventory simulator. `src/wp2/synth_regime.py` generates synthetic regime paths, mid-prices, rolling realized volatility, and detector outputs.

Gymnasium Environment Layer

`src/wp3/env.py` wraps the simulator as a Gymnasium environment. The observation vector is $[q_{\text{norm}}, \sigma_{\hat{}}, \tau, \text{regime}_L, \text{regime}_M, \text{regime}_H]$. Warmup, invalid, or disabled regime labels produce a zero one-hot vector rather than an artificial medium label.

Strategy and Training Layer

The training layer fits PPO policies, the main evaluation layer compares strategies and ablations, and the signal-informativeness layer tests whether regime labels gain value as $\sigma_{\hat{}}$ is degraded. The fixed-spread and AS baselines provide non-learning comparators.

Evaluation and Evidence Layer

Evaluation records metrics, figures, and summaries for the reported experiments. The manuscript cites the existing canonical outputs and keeps document migration separate from experiment execution.

Methodology Pipeline Illustration

Figure 3.2 summarizes the evidence pipeline at a methodological level: synthetic market paths generate the mid-price and latent regime sequence; rolling realized volatility and detector labels form the observed signals; the Gymnasium environment converts signals, inventory, time, quotes, and fills into observations and rewards; and PPO variants and analytical baselines are evaluated on held-out out-of-sample

metrics. The key implication is that signal formation, policy interaction, and evaluation remain separate and traceable stages rather than a single opaque performance step.

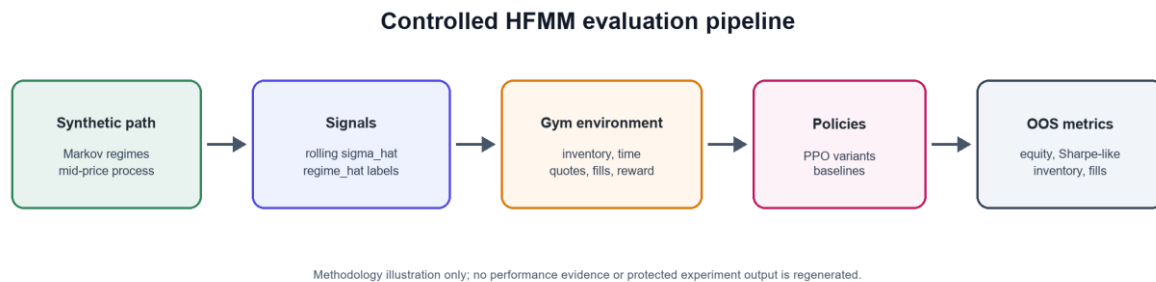


Figure 3.2. Methodology-only HFMM evaluation pipeline. Synthetic market paths produce σ_hat and $regime_hat$ signals, which feed the Gymnasium environment, PPO variants, baselines, and out-of-sample metrics. This illustration is not performance evidence.

3.4 Layered Experimental Model

The preceding architecture description is restated here as a conceptual layered experimental model.

Layer 1: Market State Generation

Generate Markov regimes and mid-price paths under controlled volatility multipliers.

Layer 2: Signal Construction

Construct σ_hat from rolling realized volatility and derive estimated regime labels.

Layer 3: Observation Encoding

Encode inventory, volatility, time-to-horizon, and optional regime one-hot channels.

Layer 4: Action and Execution

Decode PPO actions into bid/ask quote distances and sample Poisson-arrival fills.

Layer 5: Learning and Evaluation

Train PPO on the chronological train segment and evaluate deterministic policies OOS.

3.5 Algorithm Specification and Pseudocode

Plain-Language Overview

The experiment creates a synthetic market path, estimates volatility signals, trains policies on the first 70% of the path, and evaluates all strategies on the held-out 30%. Ablations modify which volatility channels the PPO policy can observe.

Synthetic Market and Regime Generation

Algorithm 1: Synthetic market and regime generation

Input: seed, transition matrix P , base volatility, volatility multipliers, n_steps

Initialize z_0 and S_0

For $t = 0 \dots n_steps - 1$:

 sample z_{t+1} from $P[z_t]$

 set $\sigma_t = \text{base_sigma} * \text{multiplier}[z_t]$

 sample mid-price increment and update S_{t+1}

Compute rolling σ_{hat} after warmup window

Calibrate thresholds on warmup data and assign regime_hat

Output: exogenous table with mid, σ_{hat} , regime_true , regime_hat

Market-Making Environment Step

Algorithm 2: Market-making environment step

Input: action (h_idx , m_idx), current simulator state, exogenous row

Decode $h = h_idx + 1$ and $m = m_idx - 2$

Set $\Delta_{bid} = \max(1, h + m)$, $\Delta_{ask} = \max(1, h - m)$

Compute fill intensities and per-step fill probabilities

Sample bid and ask fills

Update inventory and cash, including fees

Advance mid-price and compute W_{t+1}

Return observation, reward = $\Delta_{equity} - \eta * \text{inventory}^2$, done flag, info

PPO Training and OOS Evaluation Protocol

Algorithm 3: PPO training and OOS evaluation protocol

For each seed and variant:

 generate one exogenous synthetic path

 split path chronologically into train and test segments

 configure observation channels for the variant

 train PPO on the train segment for the configured timesteps

evaluate the deterministic policy on the test segment

log Sharpe-like ratio, final equity, inventory p99, fill rate, and per-regime metrics

Aggregate seed-paired statistics across variants

Experiment 5: Signal-Informativeness Sweep

Algorithm 4: Signal information-value sweep

For each degradation condition in {full, noisy, lagged, coarsened, none}:

 transform $\sigma_{\hat{}}$ according to the condition

 for each valid variant and seed:

 train PPO and evaluate OOS

Aggregate condition-variant means and paired comparisons

Test whether combined gains value as $\sigma_{\hat{}}$ is degraded

Reference Implementation Map

Table 3.2 Reference implementation map

Component	Reference file
Run lifecycle	src/run_context.py, run.py
Simulator	src/wp1/sim.py
Regime generation and detectors	src/wp2/synth_regime.py
Gymnasium environment	src/wp3/env.py
PPO training	src/wp4/job_w4_ppo.py
WP5 evaluation	src/wp5/job_w5_eval.py
WP6 sweep	src/wp6/job_w6_sweep_full.py

Table 3.2 links each conceptual layer to its reference implementation file. This mapping matters for reproducibility because the simulator, detector, environment, PPO training, and WP5/WP6 evaluation steps can be traced to specific source modules without regenerating protected evidence.

Complexity Analysis

Simulation and deterministic evaluation are linear in the number of environment steps. PPO training cost is approximately linear in total timesteps, policy-network forward/backward passes, and the number of seed-variant cells. The WP6 full sweep is expensive because it multiplies conditions, variants, and seeds; this is why its outputs are treated as canonical evidence.

The implementation prioritizes reproducibility and auditability over maximum simulation throughput. Future versions could improve runtime through vectorized environments, parallel rollouts, or lower-level simulator components for the execution loop.

3.6 Experimental Setup

Canonical Configuration Protocol

Canonical experiments are driven by JSON configs under ``config/``. Shared parameters include $mid0 = 100.0$, $tick_size = 0.01$, $dt = 0.2$, $baseline\ sigma_mid_ticks = 0.8$, $A = 5.0$, $k = 1.5$, $fee_bps = 0.2$, and $latency_steps = 1$.

Market context. The simulator is asset-class agnostic and uses normalized price units. It should be read as a stylized liquid electronic limit-order-book market rather than as a calibrated model of a specific venue such as equities, futures, crypto, or BIST. The fee and latency parameters are controlled friction assumptions used to make strategy comparisons internally consistent. They are not intended to identify a particular exchange microstructure.

Data Generation and Preprocessing

Each run generates synthetic mid-price and regime paths, computes rolling realized volatility, and assigns detector labels. The main reported pipelines use causal `rv_baseline` labels.

Train/Test Split

The main out-of-sample evaluation and signal-informativeness sweep use a chronological 70/30 split on the exogenous series. PPO trains on the first segment and is evaluated deterministically on the held-out OOS segment.

Strategy Variants

Experiment 1, the Main Out-of-Sample Evaluation, evaluates naive, AS, `ppo_aware`, and `ppo_blind`. Experiment 2, the Five-Variant Signal Ablation, and Experiment 5, the Signal-Informativeness Sweep, use `sigma_only`, `regime_only`, `combined`, `oracle_pure`, and `oracle_full`.

Detector Variants

Table 3.3 Detector variants

Detector	Role	Caveat
rv_baseline	Main causal detector	Rolling RV threshold; used in main WP4/WP5/WP6 pipelines.
rv_dwell	Auxiliary detector robustness	Offline dwell smoothing; not the main causal detector.
hmm	Robustness detector	Higher classification accuracy but no reliable PPO advantage.

Table 3.3 distinguishes rv_baseline as the main causal detector from rv_dwell and HMM as robustness variants. Comparing these roles tests whether the PPO result depends on detector choice; it does not redefine the auxiliary detectors as part of the main causal pipeline.

PPO Hyperparameters

Hyperparameter Selection Protocol

The thesis does not perform a broad hyperparameter optimization campaign. This is intentional because the central comparison is signal-channel design, not PPO tuning.

PPO hyperparameters are held fixed across PPO variants to avoid giving one information condition a tuning advantage.

$\eta = 0.001$ is the canonical inventory-penalty scale selected after the earlier inventory-penalty ablation.

No separate validation set is introduced because the study uses a fixed train/OOS evaluation protocol; this should be treated as a limitation and a future extension if full tuning is pursued.

This is a methodological choice, not a claim that the PPO settings are globally optimal.

Table 3.4 PPO hyperparameters

Hyperparameter	Canonical full-run value
total_timesteps	1,000,000
learning_rate	3e-4
n_steps	2048
batch_size	256

Hyperparameter	Canonical full-run value
n_epochs	10
gamma	0.999
clip_range	0.2
ent_coef	0.01 for full WP5/WP6 runs
eta	0.001

As seen in Table 3.4, the full-run PPO hyperparameters are held fixed across the compared policy variants. Using the same training budget and optimization settings prevents any variant from receiving a tuning advantage, although the settings are not claimed to be globally optimal.

4 Results

4.1 Experiment 1: Main Out-of-Sample Evaluation

Experiment 1, the 20-seed Main Out-of-Sample Evaluation, shows that PPO variants produce much higher risk-adjusted performance than the fixed-spread and AS baselines. AS can produce higher raw equity, but with substantially larger inventory exposure.

Table 4.1 Main OOS strategy metrics

Strategy	Sharpe-like	Final equity	inv_p99	Fill rate
ppo_aware	0.715	4.10	2.00	0.236
ppo_blind	0.740	4.42	2.05	0.232
AS	0.105	5.05	29.95	0.444
naive	0.127	4.49	21.20	0.119

Table 4.1 reports that the PPO variants have stronger risk-adjusted performance than the naive and Avellaneda-Stoikov baselines, while Avellaneda-Stoikov combines high raw equity with much larger inventory tail exposure. The comparison therefore matters as a joint performance-risk assessment rather than a ranking by equity alone.

AS has higher raw equity but carries substantially larger inventory tail risk; therefore, the main comparison is risk-adjusted performance and inventory control, not raw equity alone.

The AS implementation is used as a canonical analytical inventory-aware comparator rather than as an exhaustively optimized trading system. The PPO-versus-AS comparison establishes a reference point for learned quoting and inventory control, but the central thesis claim is the within-PPO signal-design comparison between $\sigma_{\hat{}}$, estimated labels, and oracle labels. The AS comparator is a fixed analytical baseline with a configured inventory-risk aversion γ and configured quote-clipping bounds; its high inv_p99 therefore reflects this chosen fixed comparator under the simulator rather than a defective implementation.

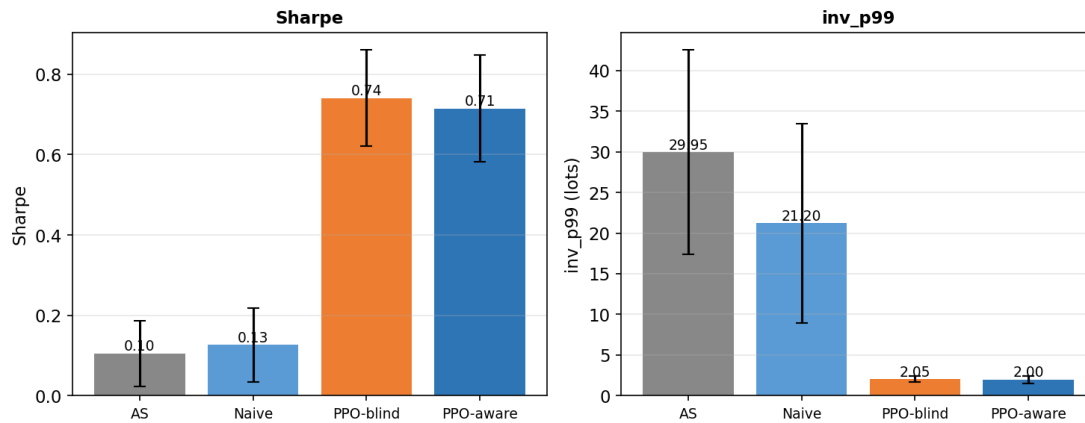


Figure 4.1. Experiment 1 main OOS results: PPO variants dominate naive and AS on risk-adjusted performance, while AS carries much larger inventory tail risk.

As shown in Figure 4.1, the PPO strategies occupy the stronger risk-adjusted region, whereas the Avellaneda-Stoikov baseline's high equity is accompanied by pronounced inventory tail risk. The visual reinforces why Sharpe-like performance and inventory control are interpreted together.

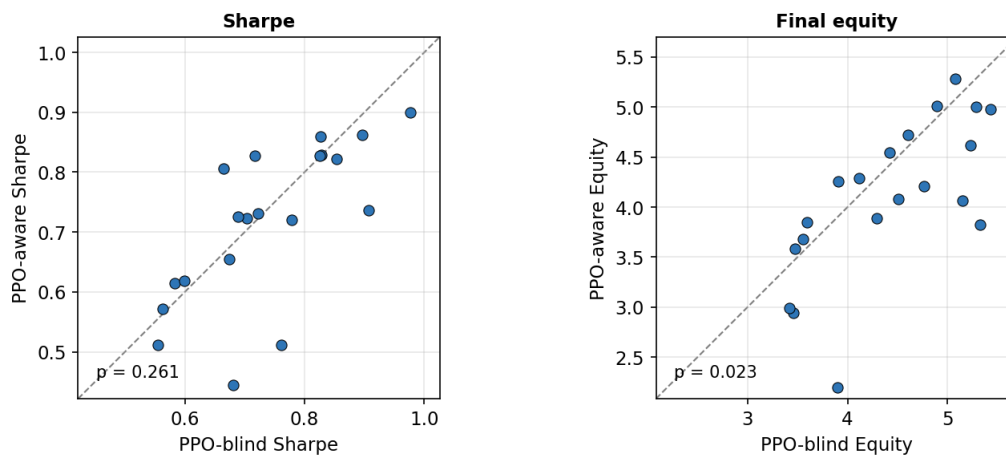


Figure 4.2. Seed-paired PPO-aware versus PPO-blind comparison: the estimated regime channel does not create a robust Sharpe-like advantage.

As shown in Figure 4.2, the aware policy does not consistently dominate the blind policy seed by seed, and the equity comparison favors the blind policy in the canonical paired test. The figure therefore provides no robust evidence that the estimated categorical regime channel improves PPO once $\sigma_{\hat{\sigma}}$ is already observed.

4.2 Experiment 2: Five-Variant Ablation

Table 4.2 Five-variant ablation metrics

Variant	Signals	Sharpe-like	Final equity	inv_p99
ppo_sigma_only	sigma_hat only	0.753	4.548	1.950
ppo_oracle_full	sigma_hat + regime_true	0.722	4.068	2.050
ppo_regime_only	regime_hat only	0.698	4.005	2.150
ppo_combined	sigma_hat + regime_hat	0.696	3.905	1.800
ppo_oracle_pure	regime_true only	0.684	3.925	1.950

As seen in Table 4.2, sigma_only has the strongest mean Sharpe-like result among the five variants, while the combined and oracle-label variants do not establish a robust label advantage. The table matters because it isolates the incremental value of categorical information from the continuous sigma_hat signal; the accompanying TOST result supports practical equivalence between sigma_only and oracle_full under the stated bound.

The label-quality objection is also bounded by the oracle_pure result. oracle_pure receives the true categorical regime label but not sigma_hat, yet it remains below sigma_only in mean Sharpe-like performance in the five-variant ablation. This suggests that the limitation is not only detector noise: in this calibration, the continuous volatility signal is more useful for quote control than the discretized regime category. The small oracle_pure versus regime_only gap in this group is well within the overlapping seed-level confidence intervals reported for these variants and is not statistically distinguishable.

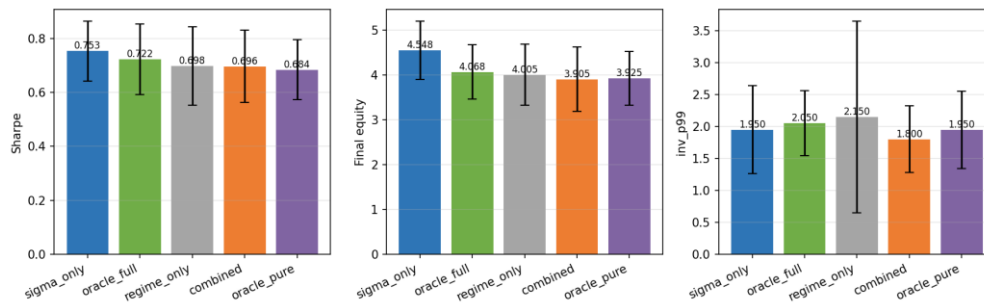


Figure 4.3. Five-variant ablation summary: *sigma_only* is the strongest mean Sharpe-like variant, and adding categorical labels does not improve it.

Figure 4.3 illustrates the same ablation pattern visually: *sigma_only* has the strongest mean Sharpe-like performance, and adding estimated or true categorical labels does not improve that benchmark robustly. This supports a bounded signal-redundancy interpretation within the tested synthetic environment.

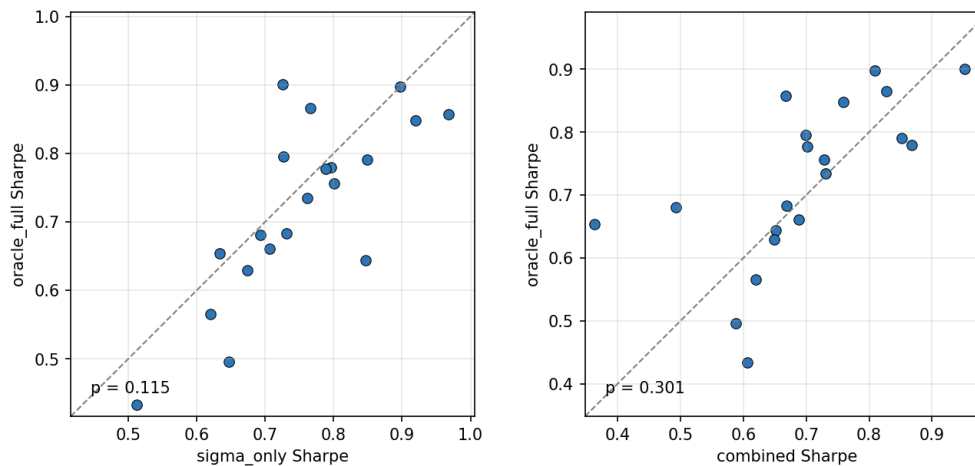


Figure 4.4. Oracle-label paired-seed comparison: even true regime labels do not reliably improve on *sigma_hat* alone.

As shown in Figure 4.4, *oracle_full* does not reliably improve on *sigma_only* across paired seeds. Because the categorical input is the true regime in this comparison, detector noise alone cannot explain the absence of a robust label advantage.

The compact statistical summary below collects the canonical paired tests used to interpret the ablation and robustness results.

Table 4.3 Compact statistical summary

Comparison	Metric or test	Canonical result	Interpretation
PPO-aware vs PPO-blind	Sharpe-like paired t-test	$p = 0.261$	No significant Sharpe improvement.
PPO-aware vs PPO-blind	Final equity paired t-test	$p = 0.023$	Favors PPO-blind.
ppo_sigma_only vs ppo_oracle_full	Sharpe-like paired t-test	$p = 0.115$	No oracle-label Sharpe improvement.
ppo_sigma_only vs ppo_oracle_full	TOST ± 0.10	$p = 0.00067$; 90% CI $[-0.001, +0.063]$	Practical equivalence supported.
HMM detector	Detector accuracy	81.8%	Higher accuracy does not create advantage.
Detector robustness	ANOVA	$F = 0.0034$, $p = 0.9966$	Detector choice does not explain result.
Regime-conditional eta	sigma_only vs combined Sharpe	$p = 0.0016$	Favors sigma_only.
Mild misspecification	sigma_only vs oracle_full Sharpe t-test	$p = 0.881$	No significant Sharpe difference.
Mild misspecification	TOST ± 0.05	$p = 0.042$	Practical equivalence supported.
ppo_oracle_full vs ppo_combined	Sharpe-like paired t-test	$p = 0.301$	No robust oracle-label advantage over the estimated-label combined variant.

The key implication of Table 4.3 is that the paired t-tests and TOST answer complementary questions: the former assess differences, while the latter assess practical equivalence within stated bounds. Together they support a bounded interpretation of no robust incremental label advantage rather than a universal result or a claim about PPO's internal mechanism.

4.3 Experiment 3: Detector Robustness

Table 4.4 Detector robustness summary

Detector	Aware vs blind conclusion	Key statistic
rv_baseline	No reliable aware Sharpe advantage	$p = 0.1142$
rv_dwell	No reliable aware Sharpe advantage	$p = 0.1095$
hmm	No reliable aware Sharpe advantage despite higher accuracy	$p = 0.0822$
ANOVA	Detector choice does not explain the result	$F = 0.0034$, $p = 0.9966$

Table 4.4 reports that rv_baseline, rv_dwell, and HMM each fail to produce a stable aware-policy advantage despite their different detection roles and accuracies. Detector

choice is therefore treated as a robustness check rather than a source of a new positive result.

The ANOVA statistic is retained as a descriptive robustness summary across detector-specific PPO-aware Sharpe values. Because the same seeds are shared across detector conditions, the primary inferential evidence remains the seed-paired detector-specific tests; the ANOVA should not be read as the sole formal test of a repeated-measures design.

The detector robustness table comes from Experiment 3, the Detector Robustness experiment (w5_detector_full), while the main OOS table in Section 4.1 comes from Experiment 1 (w5_main). Small differences in PPO-aware/blind means therefore reflect distinct canonical experiment batches, not an inconsistency.

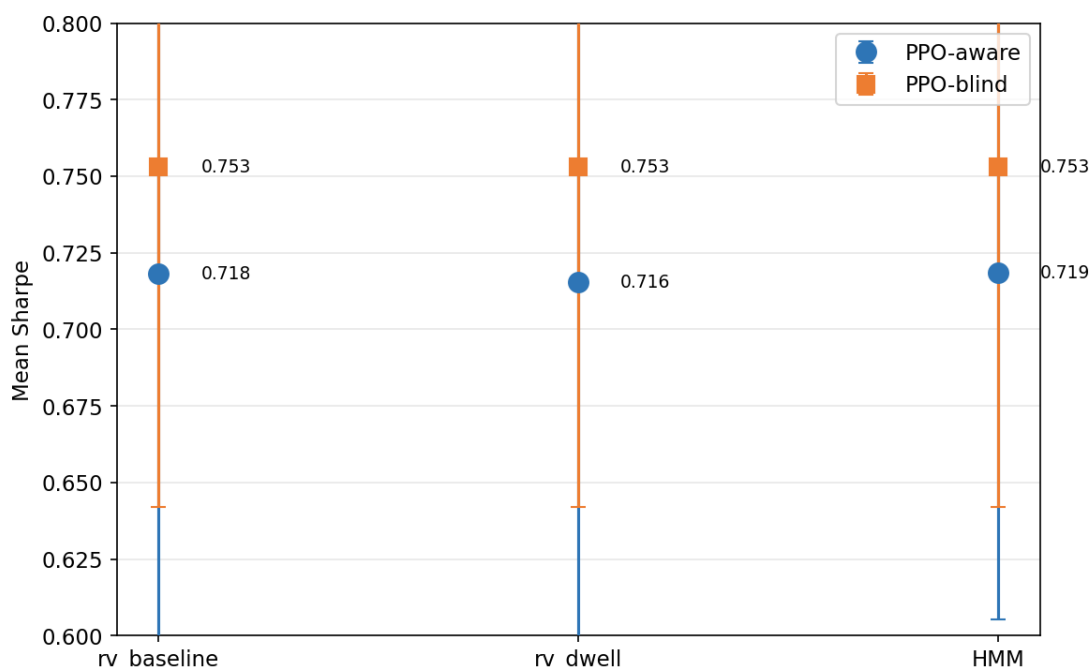


Figure 4.5. Detector robustness: *rv_baseline*, *rv_dwell*, and *HMM* all fail to create a reliable regime-aware PPO advantage.

Figure 4.5 illustrates that changing the detector does not create a consistent regime-aware Sharpe-like advantage. This visual check helps rule out the narrow explanation that the main result is caused only by the selected estimated-label method.

4.4 Experiment 4: Reward-Shaping and Misspecification Checks

The regime-conditional eta run tests whether explicit labels become useful when the reward penalizes high-volatility inventory more strongly. By design this run sets the high-volatility inventory penalty to five times the low-volatility penalty ($\eta_H = 5 \times$

eta_L). sigma_only still beats combined on Sharpe-like performance ($p = 0.0016$). Under mild regime-dependent execution misspecification, sigma_only and oracle_full remain statistically indistinguishable and practically equivalent under the specified TOST bound.

In the misspecification check, the mean difference between sigma_only and oracle_full is very small, but the ± 0.05 TOST result is close to the threshold; it should therefore be read as supportive but less decisive than the main five-variant ablation equivalence result.

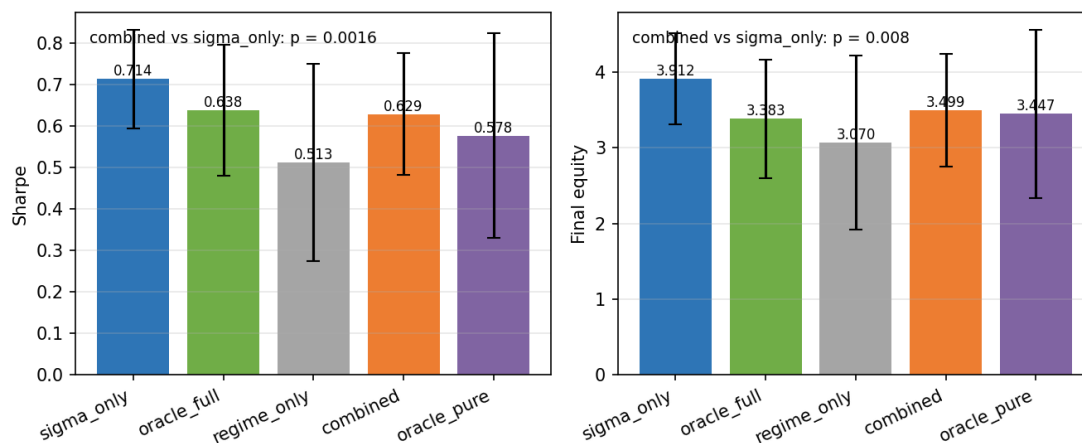


Figure 4.6. Regime-conditional eta summary: changing the reward channel by regime still favors sigma_only over combined.

As shown in Figure 4.6, making eta regime conditional still leaves sigma_only ahead of combined in the reported comparison. The result bounds the reward-shaping objection: this tested reward channel does not make the explicit label incrementally useful.

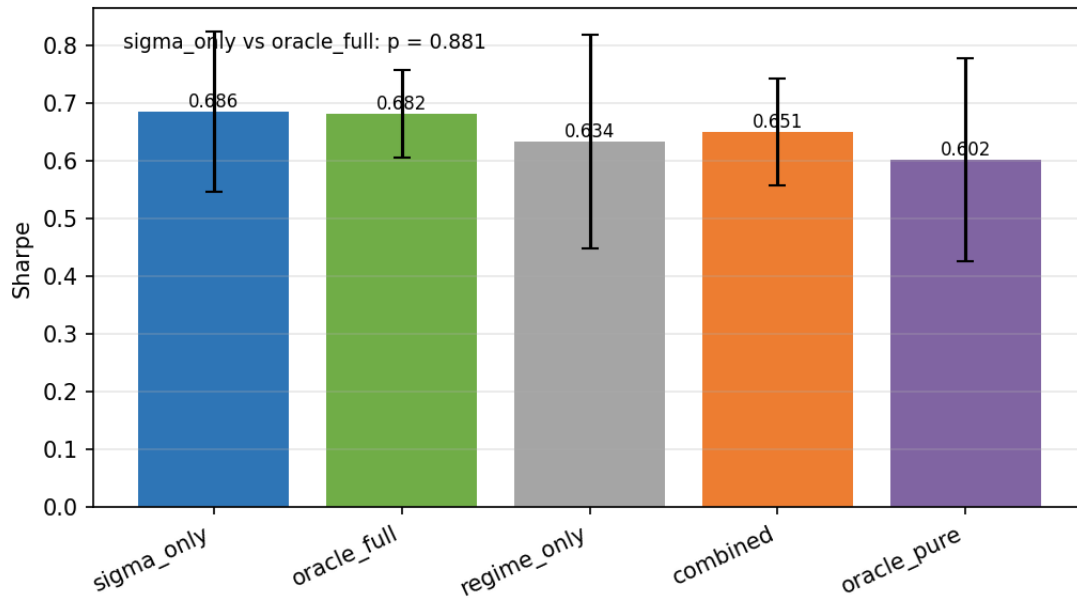


Figure 4.7. Mild model-misspecification summary: sigma_only and oracle_full remain close under regime-dependent execution parameters.

Figure 4.7 shows that sigma_only and oracle_full remain close under the mild regime-dependent execution misspecification. The implication is deliberately limited to this mild synthetic stress test; it is not a claim of robustness to arbitrary real-market misspecification.

Configuration transparency: the canonical WP3/WP4/WP5/WP6 regime setting uses sigma_mult = [0.6, 1.0, 1.8] with warmup_steps = 1000, whereas the frozen historical w5_misspec_mild run used sigma_mult = [0.5, 1.0, 2.0] with warmup_steps = 400. The mild misspecification result is retained as a frozen historical robustness check; it was not rerun or homogenized and is interpreted only under that frozen configuration.

4.5 Experiment 5: Signal-Informativeness Sweep

Table 4.5 Signal information-value sweep

condition	sigma_only	combined	regime_only	oracle_full	oracle_pure
full	0.7626 ± 0.0596	0.6898 ± 0.0607	0.6991 ± 0.0668	0.6811 ± 0.0697	0.6632 ± 0.0769
noisy	0.7563 ± 0.0616	0.7125 ± 0.0628	0.6991 ± 0.0668	0.6736 ± 0.1013	0.6632 ± 0.0769
lagged	0.7822 ± 0.0454	0.6812 ± 0.0536	0.6991 ± 0.0668	0.7264 ± 0.0686	0.6632 ± 0.0769
coarsened	0.7827 ± 0.0533	0.6908 ± 0.0827	0.6991 ± 0.0668	0.7447 ± 0.0549	0.6632 ± 0.0769
none	<i>undefined</i>	0.6991 ± 0.0668	0.6991 ± 0.0668	0.6632 ± 0.0769	0.6632 ± 0.0769

Table 4.5 reports that degrading σ_{hat} through the tested noisy, lagged, and coarsened conditions does not reveal a robust combined advantage. The sweep therefore tests, but does not support, the proposed informativeness-threshold pattern within the calibrated synthetic conditions.

Entries are mean $\pm$ 95% confidence-interval half-width across 20 seeds, not standard deviations.

Experiment 5, the Signal-Informativeness Sweep, did not support the original informativeness-threshold hypothesis within the tested calibration band. σ_{only} remained high under informative conditions, while combined was directionally below σ_{only} .

Table 4.2 and Table 4.5 come from distinct canonical experiment batches: Table 4.2 is the WP5 five-variant ablation, while Table 4.5 is the WP6 signal-informativeness sweep. Matching variant names under the full-information condition are therefore comparable in pattern but are not expected to take identical numeric values across the two batches. The confidence intervals of the middle-group variants (for example combined, regime_only, oracle_full, and oracle_pure) overlap substantially, so the exact rank ordering among those variants is sensitive to seed noise across batches. The robust pattern that holds in both tables is that σ_{only} has the highest mean Sharpe-like value and the categorical regime label does not improve on it.

The Signal-Informativeness Sweep figures should be read as a refinement of the main finding rather than a new mechanism proof: they show that the tested σ_{hat} degradation path did not reveal a regime-label advantage.

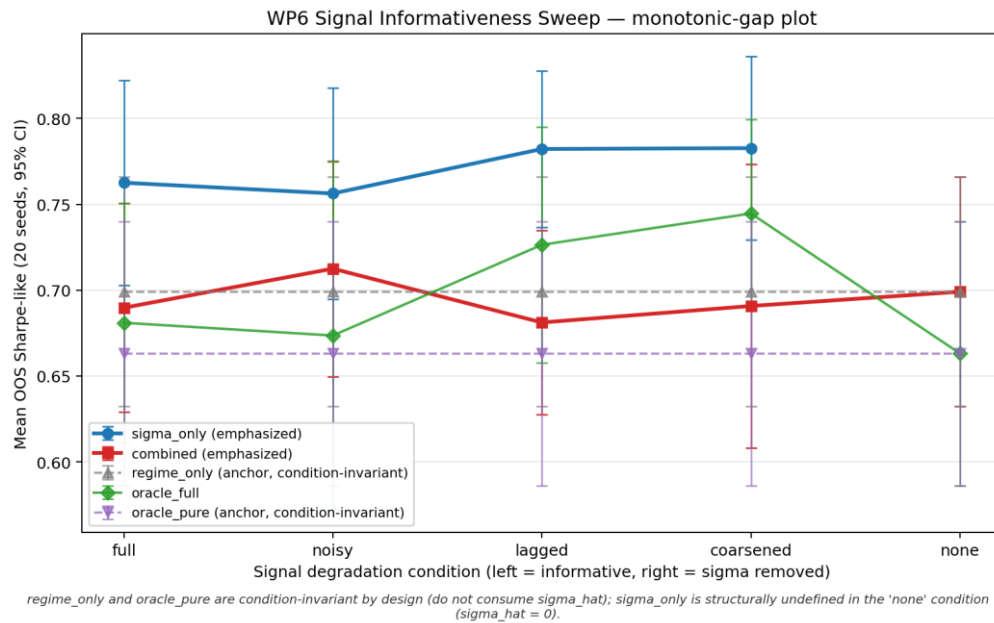


Figure 4.8. Experiment 5 monotonic-gap plot: the expected narrowing of the sigma_only versus combined gap does not appear in the tested calibration band.

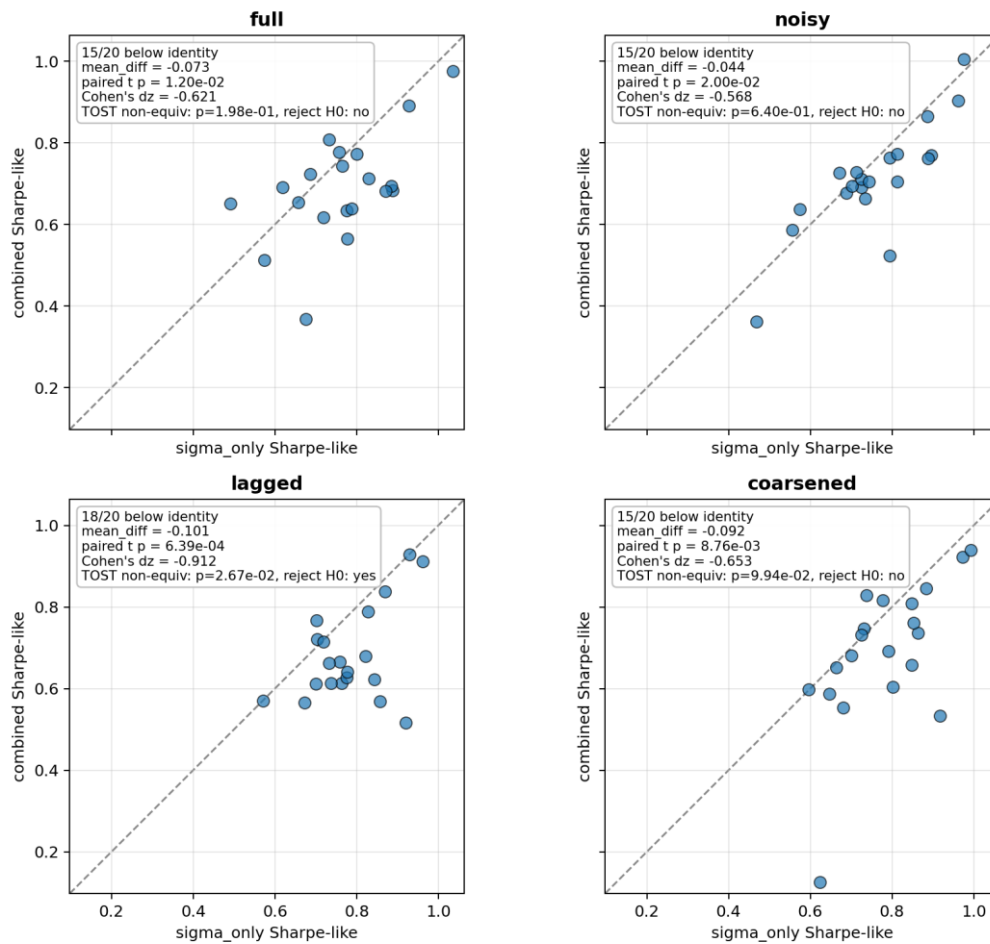
Figure 4.8 illustrates that the sigma_only-combined gap does not narrow into a robust combined advantage as the continuous signal is degraded across the tested calibration band. The figure refines the main result without identifying or proving a PPO-internal mechanism.

One possible interpretation is that the degraded continuous signal still preserves enough ordinal volatility information for quote-width adaptation, whereas the categorical label is coarser and does not add useful variation in this calibration. In noisy or lagged settings, the categorical regime channels carry non-zero content rather than being zero-filled, but this additional categorical channel content does not improve the control-relevant signal. This should be read as a performance pattern in the tested calibration band, not as proof of the PPO policy's internal representation mechanism.

The strongest degradation pattern appears in the lagged condition, where combined is materially below sigma_only (mean difference about -0.101 ; Cohen's d_z about -0.91). This suggests that, in the lagged condition, the categorical channel does not add control-relevant volatility information and is associated with lower Sharpe-like performance than sigma_only.

The paired-seed views below show whether aggregate differences are broad across seeds or driven by a few runs.

Paired-seed comparison: combined vs sigma_only across conditions

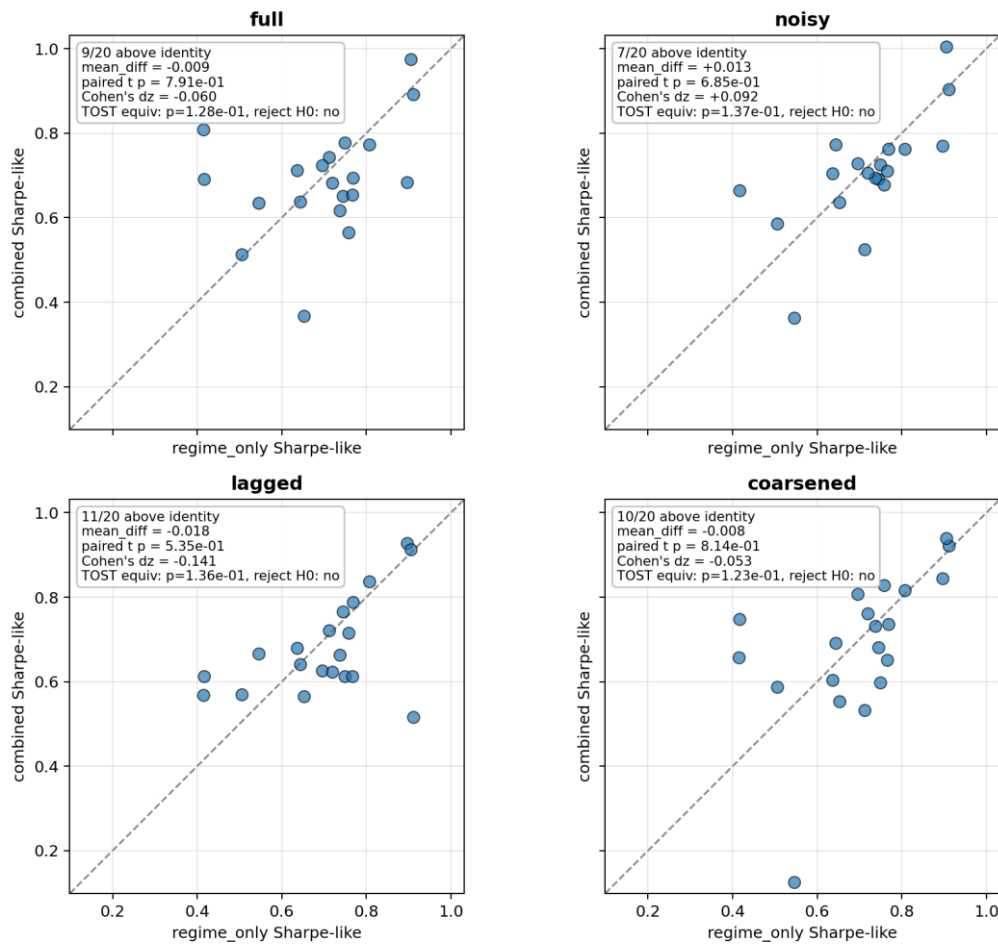


Points below identity line indicate combined underperforms sigma_only for that seed. Delta for TOST = 0.05.

Figure 4.9. Experiment 5 paired-seed combined versus sigma_only: combined is directionally below sigma_only in informative conditions.

As shown in Figure 4.9, the paired-seed comparisons place combined directionally below sigma_only across the informative conditions rather than attributing the aggregate gap to a few isolated seeds. This is paired-run performance evidence, not a universal claim about categorical inputs.

Paired-seed comparison: combined vs regime_only across conditions



Points above identity line indicate combined adds value over regime_only for that seed. Delta for TOST = 0.05.

Figure 4.10. Experiment 5 paired-seed combined versus regime_only: this diagnostic bounds how much $\sigma_{\hat{h}}$ is used inside the combined variant.

Figure 4.10 compares combined with regime_only to diagnose whether the combined policy extracts additional value from $\sigma_{\hat{h}}$ beyond the categorical channel. Read alongside Figures 4.8 and 4.9, it supports the signal-redundancy interpretation while leaving PPO's internal learning mechanism unresolved.

5 Evaluation and Discussion

5.1 Evaluation Metrics

Final Equity

Final equity is the terminal marked-to-market wealth. It is economically meaningful but not sufficient alone because high raw equity can be earned by taking large inventory risk.

Sharpe-Like Ratio

$$(16) \text{ Sharpe-like} = \text{mean}(\Delta W_t) / \text{std}(\Delta W_t) * \sqrt{1/dt}$$

Let $\Delta W_t = W_t - W_{t-1}$. The reported Sharpe-like metric is computed as $\text{mean}(\Delta W_t) / \text{std}(\Delta W_t)$, using the sample standard deviation with $\text{ddof} = 1$, multiplied by $\sqrt{1/dt}$ when the standard deviation is positive; otherwise, it is reported as 0.

Inventory Tail Risk

$$(17) \text{ inv_p99} = \text{percentile_99}(q_t)$$

Here q_t is inventory. inv_p99 is a practical tail-risk diagnostic for whether a strategy earns returns by carrying large inventory.

Fill Rate

$$(18) \text{ fill_rate} = \text{filled quote opportunities} / \text{total quote opportunities}$$

The numerator and denominator follow the logged project convention. Fill rate is diagnostic rather than the primary thesis metric.

Paired t-Tests

$$(19) t = \text{mean}(d_i) / (\text{sd}(d_i) / \sqrt{n})$$

Here d_i is the paired seed-level difference between two strategies, $\text{sd}(d_i)$ is its sample standard deviation, and n is the number of paired seeds.

Because the manuscript reports several paired comparisons, isolated marginal p-values should be read descriptively rather than as a stand-alone familywise discovery claim. The main conclusion is based on the repeated pattern across the main OOS evaluation, detector robustness, oracle ablations, reward-shaping, misspecification, and signal-informativeness diagnostics.

TOST Equivalence Tests

TOST evaluates whether the plausible range of a paired difference lies inside a pre-specified practical-equivalence band. At $\alpha = 0.05$, the corresponding equivalence reading uses the 90% confidence interval.

The equivalence bounds should be read as practical smallest-effect-size thresholds for the Sharpe-like metric, not as universal constants. A wider specified TOST bound of ± 0.10 is used for the main five-variant ablation; a narrower ± 0.05 bound is used for tighter robustness checks. The ± 0.10 band was specified in advance as the practical smallest-effect-size threshold for the Sharpe-like metric in the main five-variant ablation; relative to a mean Sharpe-like level around 0.75, a 0.10 difference is roughly a 13% relative deviation, which is treated as practically negligible in this market-making context. The band choice is not outcome-determining, because equivalence still held under the tighter ± 0.05 band in the mild-misspecification check ($p = 0.042$), although that margin is narrower and is therefore read as supportive rather than decisive. TOST is not treated as a stand-alone proof; it is reported alongside the paired t-tests, the confidence intervals, and the directional seed-level evidence.

5.2 Main Interpretation: Signal Redundancy

In the tested controlled synthetic HFMM environment, explicit categorical volatility-regime labels do not provide robust incremental value once $\sigma_{\hat{}}$ is already observed by the PPO policy. The results are consistent with signal redundancy.

The interpretation is that the continuous realized-volatility proxy already supplies the economically useful volatility information needed by the policy in this environment. The categorical label may be coarser or redundant when presented alongside $\sigma_{\hat{}}$.

The evidence should therefore be read slightly more precisely than pure redundancy. If a categorical label only repeated the useful information in $\sigma_{\hat{}}$, combined and σ_{only} would be expected to be approximately indistinguishable. In several experiments, however, combined is directionally below σ_{only} . This pattern is consistent with signal redundancy plus mild categorical-channel degradation: the label may be coarser than the continuous volatility estimate, may contribute an additional active categorical signal channel, or may interact with policy learning without adding control-relevant information. This is an interpretation of the observed performance pattern, not a proof of the PPO policy's internal mechanism.

5.3 Why the Result Is Not a Weak Null

The conclusion does not rest only on $p > 0.05$. It is supported by detector robustness, oracle labels, TOST equivalence, reward-channel checks, mild misspecification, and signal-degradation tests.

5.4 Synthetic-Market Boundary

The result is bounded to the tested synthetic HFMM simulator. The simulator deliberately isolates quote distance, Poisson-arrival fills, fees, latency, inventory, and volatility regimes so that the incremental information value of the regime channel can be tested cleanly.

Real limit-order books include additional mechanisms that are outside the main simulator: queue position, execution priority, adverse selection, order-book imbalance, richer latency effects, clustered or self-exciting order arrivals, non-stationary liquidity, and exchange-specific transaction-cost and microstructure rules. These mechanisms can change both the value of volatility information and the way a categorical regime label interacts with execution risk.

For that reason, the thesis should be read as controlled synthetic evidence, not as a real-market external validity claim. Richer LOB simulators, Hawkes-process LOB models, queue-reactive execution, and real-data evaluation are natural future work and could change the incremental value of regime labels.

The out-of-sample protocol should also be interpreted carefully. It is a chronological 70/30 temporal hold-out within the same synthetic data-generating process, not a distributional-shift test across fundamentally different markets. Multiple seeds improve robustness to simulation randomness, but they do not establish external validity beyond the controlled synthetic environment.

5.5 Risks to Interpretation

Table 5.1 Risks to interpretation

Risk	Mitigation
Overclaiming mechanism	Use signal-redundancy and categorical-channel degradation as interpretations, not proofs of PPO internals.
Detector causality confusion	State that <code>rv_baseline</code> is the main causal detector and <code>rv_dwell</code> is auxiliary/offline.
Synthetic external validity	Frame all claims as controlled synthetic-market claims.
Template over-compression	Keep the literature and evidence spine, but map it into template headings.
Mid-price mark-to-market assumption	Report it as a simulator evaluation convention; real liquidation would require spread/depth/queue/adverse-selection considerations.

Table 5.1 converts the main interpretation risks into defense-safe mitigation statements. Its role is to keep claims bounded to the controlled synthetic HFMM setting, distinguish robustness checks from mechanism proofs, and make the limitations explicit rather than treating them as afterthoughts.

5.6 Future Work

Evaluate stronger regime-dependent execution misspecification and richer non-stationary liquidity settings.

Test richer limit-order-book simulators with queue position, order-book imbalance, latency priority, adverse selection, and self-exciting order flow.

Use Hawkes-process LOB models and real-data evaluation in future work, not as current claims of this thesis.

Run representation-level diagnostics only as future mechanism work.

Explore alternative observation encodings and policy architectures.

Containerized environment specifications and external CI checks are natural reproducibility extensions for a later submission or public-release version.

6 Conclusion and Outlook

This thesis investigated whether explicit categorical volatility-regime labels improve PPO-based high-frequency market making when the policy already observes a continuous realized-volatility estimate. The study was deliberately framed as an information-design question rather than as a search for a new reinforcement-learning algorithm or a production-ready trading system. Within the controlled synthetic HFMM simulator, PPO learns strong risk-adjusted quoting behavior relative to the fixed-spread and Avellaneda-Stoikov baselines. The learned policies are able to adapt quote width and skew while keeping inventory exposure substantially more controlled than the analytical and naive comparators.

The central result is that explicit categorical volatility-regime labels do not provide robust incremental value once σ_{hat} is already included in the PPO observation space. This conclusion is supported by the canonical evidence chain: the main out-of-sample evaluation, the five-variant ablation, detector robustness checks, oracle-label comparisons, regime-conditional reward shaping, mild model misspecification, and the signal-informativeness sweep. The five-variant ablation is especially important because σ_{only} achieves the strongest mean Sharpe-like performance, while $\text{oracle}_{\text{full}}$ does not deliver a statistically significant or practically meaningful improvement over the continuous volatility signal alone. This pattern indicates that the categorical regime label mostly repackages information already carried by the realized-volatility proxy.

The result should be interpreted carefully. It does not imply that volatility regimes are economically irrelevant, nor does it prove that PPO internally ignores regime information. Instead, the evidence supports a narrower claim: in the tested synthetic environment and observation design, the continuous volatility signal is already sufficiently informative for the quoting problem, and adding a coarse categorical label does not improve the policy in a robust way. In some experiments, the combined label-and-sigma variants are even directionally below σ_{only} , which is consistent with signal redundancy together with possible categorical-channel degradation, the presence of an additional active categorical signal channel, or coarser categorical representation.

The thesis also clarifies the boundary of the evidence. The simulator isolates the mechanisms needed for a controlled comparison: Markov volatility regimes, rolling realized-volatility detection, Poisson-arrival fills, fees, latency, inventory, and quote-distance decisions. This makes the incremental information value of regime labels testable, but it also limits external validity. Real limit-order books include queue position, adverse selection, order-book imbalance, clustered order flow, exchange-

specific costs, richer latency effects, and non-stationary liquidity. These mechanisms may change the value of categorical regime information, especially if the label captures market structure not already summarized by realized volatility.

Overall, the strongest defensible conclusion is conditional and synthetic-market bounded. PPO is effective as a learned quoting policy in the controlled HFMM setting, but explicit categorical volatility-regime labels do not add robust decision-relevant information beyond $\sigma_{\hat{}}$ in this experiment. Future work should therefore focus less on simply adding regime labels and more on richer sources of non-redundant state information: order-book imbalance, queue-reactive execution, Hawkes-style order-flow dynamics, stronger regime-dependent execution frictions, and real-data evaluation. Containerized environments, external CI checks, and representation-level diagnostics would also strengthen future reproducibility and mechanism analysis.

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Appendix A Reproducibility & Evidence Integrity

This appendix records reproducibility context in abbreviated form. It is included for review and future reproduction, not as part of this template migration execution.

A.1 Software Environment

The implementation uses Python, JSON configuration files, fixed random seeds, chronological train/test splits, per-run configuration snapshots, stable metric schemas, and resume validation for long signal-informativeness sweeps.

A.2 Reproducibility Checklist

Table A.1 Reproducibility checklist

Checklist item	Verified value	Living/frozen source
Frozen baseline	Tag thesis-v29-frozen; commit 9681faa	FREEZE_METADATA.md, "Freeze coordinates"
Run-ID format	YYYYMMDD-HHMMSS_seed<S>_<TAG>_<COMMIT>	AGENTS.md and CLAUDE.md, "Run ID format"
WP5 main seeds	Seeds 1-20, derived from the explicit seed array	config/w5_main.json, wp5.seeds
WP6 full seeds	Seeds 42-61, derived from the explicit seed array	config/w6_sweep_full.json, sweep.seeds
Train/test split	70% training / 30% OOS, chronological	config/w5_main.json, wp5.train_frac = 0.7; AGENTS.md and CLAUDE.md, "Train/test split"; current HECTOR manuscript for the existing chronological-split wording
No-rerun policy	The document migration did not execute PPO training, WP5/WP6 evaluation, detector experiments, ablations, or signal sweeps.	Current HECTOR manuscript, existing Appendix A text; AGENTS.md and CLAUDE.md, documentation-only no-rerun guardrail

A.3 Protected-Evidence Hash Register

These are the canonical protected artifacts verified 4/4 SHA256 MATCH according to EVIDENCE_MANIFEST.md.

Displayed hashes are abbreviated to the first 12 and last 4 characters for readability; full SHA256 values are recorded in EVIDENCE_MANIFEST.md.

Table A.2 Protected-evidence hash register

Artifact path	SHA256 (abbreviated)	Status
results/metrics_detector_compare.csv	28E7AD40BB47...06ED	MATCH

Artifact path	SHA256 (abbreviated)	Status
docs/internal/wp6_sweep_full/summary_condition_varia nt.csv	6DD627E81637...C6EA	MATCH
docs/internal/wp6_sweep_full/summary_paired_combine d_vs_sigma.csv	4BABCAAACE1D...B796	MATCH
docs/internal/wp6_sweep_full/summary_paired_combine d_vs_regime.csv	2087FEFBE5DC...E8F9	MATCH

A.4 Run/Command & Entry-Point Map

Reproducibility commands:

```
python -m venv .venv
source .venv/Scripts/activate
pip install -r requirements.txt
ruff check src/
pytest -q
```

Command sources: AGENTS.md for environment creation, activation, dependency installation, and linting; existing Appendix A text for pytest -q.

Never execute scripts/eval_only_seed1to7.py — it overwrites a protected figure-input CSV and breaks the thesis_29 provenance chain.

Source: scripts/eval_only_seed1to7.py warning and AGENTS.md protected-output guardrail; provenance context is recorded in FREEZE_METADATA.md and EVIDENCE_MANIFEST.md.

Sources: [C] docs/internal/config_snapshot_all.md (active inventory and job values); [A] AGENTS.md (experiment descriptions and rerun guardrails).

Table A.3 Active configuration, job, and rerun-status map

Config	WP / job and experiment	Rerun status	Source
config/base.json	WP0 / no job key - base smoke configuration	NOT RUN - document migration only	[C]; [A]
config/w1_as_baseline.json	WP1 / w1_as_baseline - single AS episode	NOT RUN - document migration only	[C]; [A]
config/w1_compare.json	WP1 / w1_compare - naive vs AS comparison	NOT RUN - document migration only	[C]; [A]
config/w1_naive_sweep.json	WP1 / w1_naive_sweep - fixed half-spread sweep	NOT RUN - document migration only	[C]; [A]
config/w2_synth.json	WP2 / w2_synth - synthetic regime generation and detection	NOT RUN - document migration only	[C]; [A]
config/w3_sanity.json	WP3 / w3_sanity - regime-aware sanity check	NOT RUN - document migration only	[C]; [A]
config/w3_sanity_both.json	WP3 / w3_sanity - aware and blind sanity checks	NOT RUN - document migration only	[C]; [A]
config/w4_ppo.json	WP4 / w4_ppo - PPO training	DO NOT RERUN - documentation-only work	[C]; [A]

Config	WP / job and experiment	Rerun status	Source
config/w5_ablation_eta.json	WP5 / w5_ablation_eta - eta ablation	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_ablation_skew.json	WP5 / w5_ablation_skew - skew-penalty ablation	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_detector_full.json	WP5 / w5_detector_compare - full detector-robustness evaluation	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_detector_pilot.json	WP5 / w5_detector_compare - pilot detector-robustness evaluation	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_eta_regime.json	WP5 / w5_eval - regime-conditional eta experiment	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_eval.json	WP5 / w5_eval - OOS evaluation	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_main.json	WP5 / w5_eval - main OOS evaluation	DO NOT RERUN - documentation-only work	[C]; [A]
config/w5_misspec_mild.json	WP5 / w5_eval - mild model-misspecification experiment	DO NOT RERUN - documentation-only work	[C]; [A]
config/w55_audit.json	WP5.5 / w55_audit - offline signal-degradation audit	NOT RUN - document migration only	[C]; [A]
config/w55_calibration.json	WP5.5 / w55_calibration - signal-degradation calibration	NOT RUN - document migration only	[C]; [A]
config/w55_runtime.json	WP5.5 / w55_runtime - runtime profiling	NOT RUN - document migration only	[C]; [A]
config/w6_sweep_full.json	WP6 / w6_sweep_full - full signal-informativeness sweep	DO NOT RERUN - documentation-only work	[C]; [A]
config/w6_sweep_pilot.json	WP6 / w6_sweep_pilot - pilot signal-informativeness sweep	DO NOT RERUN - documentation-only work	[C]; [A]

“NOT RUN” records that a configuration was not executed during this document migration; “DO NOT RERUN” identifies configurations that must not be re-executed because doing so would disturb frozen evidence.

A.5 Configuration Inventory

Source: docs/internal/config_snapshot_all.md. This compact inventory lists the 21 active configuration snapshots, their literal job fields, and selected role-defining values copied as recorded.

Table A.4 Active configuration inventory

Config	Literal job	Compact role / key snapshot values
config/base.json	[no job key]	WP0 smoke/base; run_tag=wp0-smoke; n_steps=200
config/w1_as_baseline.json	w1_as_baseline	AS baseline; episode.n_steps=8000; as.gamma=0.01
config/w1_compare.json	w1_compare	Naive/AS comparison; sweep.half_spreads_ticks=[1, 2, 3, 4, 5]
config/w1_naive_sweep.json	w1_naive_sweep	Fixed half-spread sweep; sweep.half_spreads_ticks=[1, 2, 3, 4, 5]

Config	Literal job	Compact role / key snapshot values
config/w2_synth.json	w2_synth	Generator-stage regime configuration; sigma_mult=[0.5, 1.0, 2.0]; warmup_steps=1000
config/w3_sanity.json	w3_sanity	Sanity configuration; wp3.use_regime=true; wp3.eta=0.01
config/w3_sanity_both.json	w3_sanity	Dual sanity configuration; run_tag=wp3-sanity-both; wp3.use_regime=false
config/w4_ppo.json	w4_ppo	PPO configuration; wp4.total_timesteps=200000
config/w5_ablation_eta.json	w5_ablation_eta	Eta ablation; wp5.eta_values=[0.0001, 0.001, 0.01]
config/w5_ablation_skew.json	w5_ablation_skew	Skew-penalty ablation; wp5.skew_c_values=[0.0001]
config/w5_detector_full.json	w5_detector_compare	Detector comparison (full); run_tag=wp5-detector-full; wp4.total_timesteps=1000000
config/w5_detector_pilot.json	w5_detector_compare	Detector comparison (pilot); run_tag=wp5-detector-pilot; wp5.seeds=[1, 2, 3]
config/w5_eta_regime.json	w5_eval	Regime-conditional eta; wp3.eta_regime={"L": 0.0005, "M": 0.001, "H": 0.0025}
config/w5_eval.json	w5_eval	OOS evaluation; wp5.seeds=[11, 22, 33]; wp4.total_timesteps=200000
config/w5_main.json	w5_eval	Main OOS configuration; wp5.train_frac=0.7; wp4.total_timesteps=1000000
config/w5_misspec_mild.json	w5_eval	Frozen historical robustness configuration; misspec.enabled=true; sigma_mult=[0.5, 1.0, 2.0]; warmup_steps=400
config/w55_audit.json	w55_audit	Offline signal audit; audit.noise_std=auto; lag_k_steps=5; n_bins=5
config/w55_calibration.json	w55_calibration	Degradation calibration; calibration.n_seeds=5; alpha_values=[0.0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6]
config/w55_runtime.json	w55_runtime	Runtime profiling only (not a reported experiment); market.dt=1.0; market.sigma_mid_ticks=0.5; benchmark.devices=["cpu", "cuda"]
config/w6_sweep_full.json	w6_sweep_full	Full signal-informativeness sweep; sweep.timesteps=1000000; conditions=["full", "noisy", "lagged", "coarsened", "none"]
config/w6_sweep_pilot.json	w6_sweep_pilot	Pilot signal-informativeness sweep; sweep.timesteps=200000; sweep.seeds=[42, 43, 44]

Critical configuration note: the canonical main WP3/WP4/WP5/WP6 regime setting uses sigma_mult = [0.6, 1.0, 1.8] with warmup_steps = 1000. config/w2_synth.json contains sigma_mult = [0.5, 1.0, 2.0] as an earlier generator-stage value. config/w5_misspec_mild.json is a frozen historical robustness run with sigma_mult = [0.5, 1.0, 2.0] and warmup_steps = 400. config/w55_runtime.json is a runtime-profiling configuration, not a reported experiment.

Appendix B Supplementary Evidence & Diagnostics

These diagnostics support interpretation and provenance; they do not replace the canonical frozen evidence chain used for the main claims.

B.1 Signal-Informativeness Sweep Summary

Table B.1 Signal-informativeness condition/variant summary

condition	sigma_only	combined	regime_only	oracle_full	oracle_pure
full	0.763 [0.703, 0.822]	0.690 [0.629, 0.751]	0.699 [0.632, 0.766]	0.681 [0.611, 0.751]	0.663 [0.586, 0.740]
noisy	0.756 [0.695, 0.818]	0.713 [0.650, 0.775]	0.699 [0.632, 0.766]	0.674 [0.572, 0.775]	0.663 [0.586, 0.740]
lagged	0.782 [0.737, 0.828]	0.681 [0.628, 0.735]	0.699 [0.632, 0.766]	0.726 [0.658, 0.795]	0.663 [0.586, 0.740]
coarsened	0.783 [0.729, 0.836]	0.691 [0.608, 0.773]	0.699 [0.632, 0.766]	0.745 [0.690, 0.800]	0.663 [0.586, 0.740]
none	not defined	0.699 [0.632, 0.766]	0.699 [0.632, 0.766]	0.663 [0.586, 0.740]	0.663 [0.586, 0.740]

Note: Values are copied from summary_condition_variant.csv and display-rounded only; no statistics were recomputed.

B.2 Paired-Seed Diagnostic Summaries

Table B.2 Combined versus sigma_only paired diagnostics

condition	n_below/n_total	mean_diff	paired p	TOST p	TOST reject
full	15/20	-0.073	0.012	0.198	No
noisy	15/20	-0.044	0.020	0.640	No
lagged	18/20	-0.101	0.00064	0.027	Yes
coarsened	15/20	-0.092	0.0088	0.099	No

Note: Values are copied from summary_paired_combined_vs_sigma.csv and display-rounded only; mean_diff is combined - sigma_only; no statistics were recomputed.

Table B.3 Combined versus regime_only paired diagnostics

condition	n_above/n_total	mean_diff	paired p	TOST p	TOST reject
full	9/20	-0.009	0.791	0.128	No
noisy	7/20	0.013	0.685	0.137	No
lagged	11/20	-0.018	0.535	0.136	No
coarsened	10/20	-0.008	0.814	0.123	No

Note: Values are copied from summary_paired_combined_vs_regime.csv and display-rounded only; mean_diff is combined - regime_only; no statistics were recomputed.

B.3 Existing Diagnostic Figures

The following figures use already-rendered PNG artifacts and were embedded without regeneration.

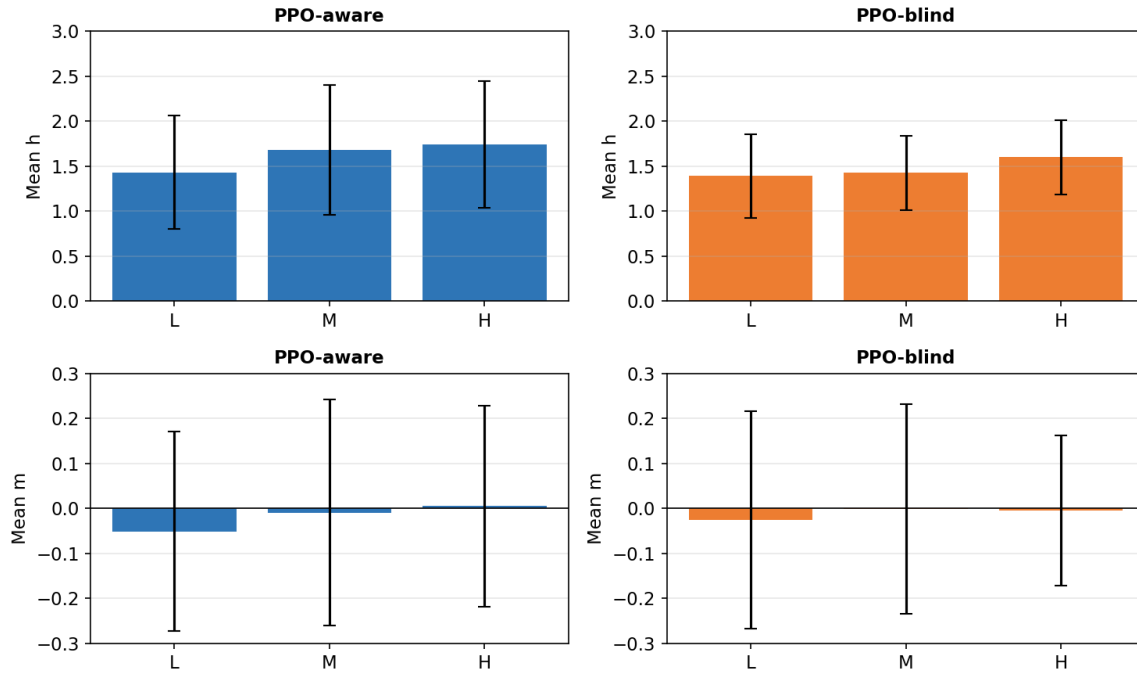


Figure B.1. Regime-wise action distribution diagnostic.

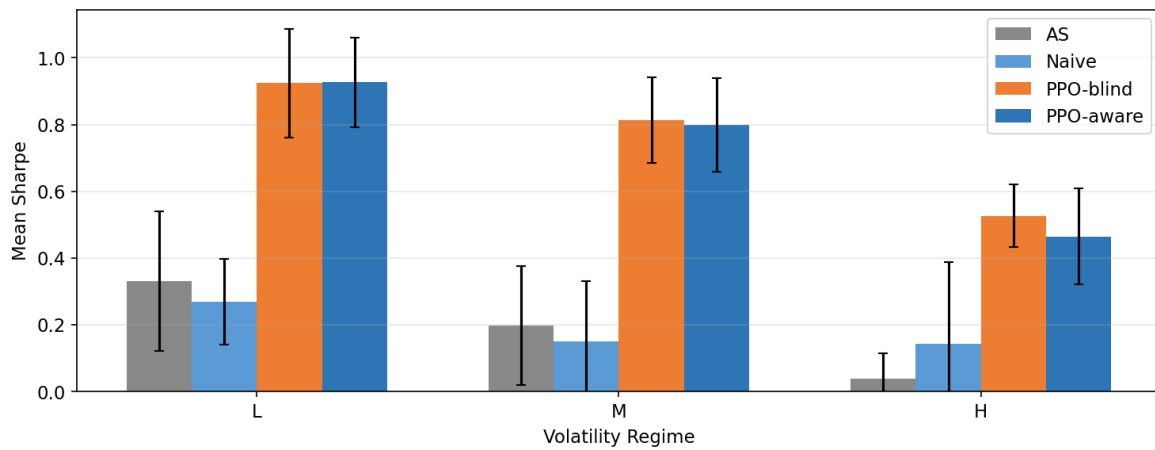


Figure B.2. Regime-wise Sharpe diagnostic.

Appendix C Author Contribution Statement

This thesis is submitted as a single-author MSc thesis by Onur Emiroglu. The reported implementation, experiments, documentation, and interpretation are organized in the accompanying project workspace and manuscript artifacts.